

Select The Right ENERGY HARVESTER For Your Next Embedded Project

Stand-alone ambient energy harvesting has become an important issue in the world of Internet of Things and medical implants, where batteries cannot be relied upon for long-time use. Here are some solutions that are emerging. But you need to select the right one for your application

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Advancement in low power electronics has paved the way to make a truly stand-alone ultra-low powered embedded product. This feat is made possible by ambient energy scavenging or, in other words, energy harvesting. Energy harvesting is a method of harnessing and converting the ambient energy into electrical energy for immediate use or to be stored for later usage. In short, energy harvesting makes it possible to realise the goal of a truly stand-alone system.

Selecting the right energy source is important for a design engineer as it dictates the run time, cost of operation, and even the size of the product. Battery needs to be replenished and any sort of non-renewable energy generation from any source like gas or solid fuel requires periodic refuelling, which is not possible for off-grid sensors and devices placed at remote locations. But the ambient environment contains energy which can be

harnessed to run the low-power electronics for an extended period without any human intervention. Thus, the need for ambient energy scavenging arises.

Energy harvesters are devices that can convert different forms of ambient energy into electrical energy, thus eliminating the requirement of a battery or extending the battery life of an ultra-low-powered system. It is a combination of transducers and power conditioning circuits. The energy can either be directly used to power a device, or the energy can be stored in a storage device for later use. The block diagram of an energy harvester is shown in Fig. 1.

The conversion circuit of an energy harvester usually incorporates a booster which increases the very low extractable voltage into higher voltage output capable of charging a cell or for directly powering a circuit. Some models also incorporate maximum power point tracking (MPPT)

algorithms to maintain the optimum current to voltage ratio so that it can capture the maximum amount of energy.

The main task of an

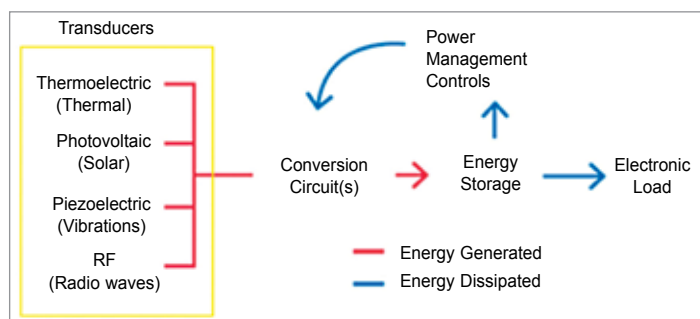


Fig. 1: Energy harvester block diagram

TABLE 1
POWER DENSITIES
OF AMBIENT ENERGY
SOURCES

Energy source	Power density
Solar	10 μ W - 25mW
Thermal	20 μ W - 10mW
RF	0.01 μ W - 0.1 μ W
Vibration	4 μ W - 100 μ W